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Prediction of mechanical properties of high entropy alloys based on machine learning

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Abstract

In recent years, the ideal- properties (young's modulus, yield strength, toughness) and advanced application potential of high-entropy alloys (HEAs) have attracted numerous researchers. However, due to their unique structure and multiple structural combinations, it is challenging to explore the impact of various factors on their mechanical performance solely through experiments. This study considers the concentrations of five high-entropy alloy atoms and working temperature as input parameters. Molecular dynamics (MD) simulations and machine learning (ML) algorithms are employed to predict the tensile properties of FeNiCrCoCu HEAs, including Young's modulus (E) and toughness (uT). A dataset of 1000 HEAs is generated through MD simulations, and feature selection is conducted using principal component analysis and Spearman correlation analysis. XGBoost, RF, DT, LGBoost, and AdaBoost are utilized to predict the mechanical properties of HEAs, comparing the impact of the two feature selection methods on prediction outcomes. During ML model training, 10-fold cross-validation and grid search are employed to obtain the best models and parameters. Root mean squard error ($RMSE$), coefficient of determination (R^2), mean absolute error (MAE) and relative absolute error (RAE) are used as evaluation metrics. Results indicate that Spearman correlation analysis for feature selection outperforms principal component analysis, and XGBoost demonstrates superior predictive performance for the mechanical properties of HEAs compared to other models. Predictions for E are more accurate than those for uT, with R^2 exceeding 0.9 for four out of the five ML models. This work may provide a new feature selection method for studying the mechanical properties of HEAs through ML. In the future, this method can be applied to other research areas of HEAs compositions, providing theoretical support for experiments. It can then be further applied to critical fields such as biomedical and aerospace industries.

1. Introduction

Since the introduction of the concept of High-Entropy Alloys (HEAs) by Yeh *et al* in 2004 [1], these materials have been extensively studied due to their superior mechanical properties. These include excellent ductility and fracture toughness at low temperatures, high-temperature strength, and corrosion resistance. Consequently, HEAs hold significant potential for applications in various critical fields such as the harsh corrosive environments of the ocean, the intense pressures of deep-sea operations, aerospace, and nuclear energy [2–5]. Research on HEAs traditionally involves experimental synthesis or computational simulation methods. Traditional experimental methods face challenges in observing microstructural evolution, particularly in HEAs composed of five or more elements. Some researchers have addressed this issue by employing molecular dynamics (MD) simulations to study the performance of HEAs [6–9]. Current research on HEAs often relies on experimental synthesis or computational simulations. While traditional studies have demonstrated the enormous potential of HEAs, experimental methods are constrained by time and spatial scales. Experimental synthesis of multi-component HEAs requires significant time and material costs, while computational

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simulations are time-consuming and have inherent limitations. Moreover, for researchers in higher education institutions, high-end experimental equipment is expensive and has high maintenance costs, which are difficult for many institutions to afford. This results in limited equipment resources to meet the experimental needs of researchers.

Current research on HEAs often relies on experimental synthesis or computational simulation methods, but these are limited by the time and spatial scales of experimental approaches. Experiments require a significant amount of time and material costs, while computational simulations are time-consuming, labor-intensive, and have certain limitations [10]. With the rapid development of artificial intelligence, machine learning (ML) is gradually demonstrating revolutionary advantages in the field of materials science. Data-driven approaches significantly accelerate the research and development process, reducing time and computational costs [11]. Researchers have begun to employ a combination of ML with MD simulations and experimental methods to study HEAs [12, 13]. Cheng Wen *et al* [14] formulated a materials design strategy that integrates an ML surrogate model with experimental design algorithms to explore HEAs with high hardness in an Al-Co-Cr-Cu-Fe-Ni system. Omarelfarouq Elgack *et al* [15] utilized MD simulations and ML algorithms considering factors such as atomic concentrations, grain size, working temperature, and strain rate to predict the Young's modulus, yield strength, and ultimate tensile strength of FeNiCrCoCu HEAs. Lei Jiang *et al* [16] optimized the composition of AlFeCuSiMg HEAs based on elastic modulus as a prediction index using ML and MD approaches. Jingui Yu *et al* [17] combined ML with MD to predict the mechanical properties of FeNiCrCoCu, finding that the error in predicted tensile strength via ML was within 0.5% of that obtained from MD simulations.

In this study, a combined approach of ML and MD was employed. MD simulations were conducted to simulate the tensile process of FeNiCrCoCu HEAs, while ML investigated the mechanical properties of these alloys. A comparison was made of two feature selection methods, and a novel predictive framework was proposed for the FeNiCrCoCu system. Initially, a dataset was generated using MD with different input parameters: atomic concentration (C) of constituent elements and operating temperature (T), which were necessary for ML. Subsequently, Principal Component Analysis (PCA) and Spearman correlation coefficient (PCC) were applied for feature selection. Five ML models—Extreme Gradient Boosting (XGBoost), AdaBoost, Random Forest (RF), Light Gradient Boosting Machine (LGBoost), and Decision Trees (DT)—were then employed to predict two mechanical properties of FeNiCrCoCu HEAs: Young's modulus (E) and toughness (uT). During model training, techniques such as 10-fold cross-validation and grid search were utilized for parameter tuning. Finally, the performance of the models on the test set was evaluated after feature selection using PCC and PCA methods, with results compared and models interpreted using Shapley Additive Explanations (SHAP) techniques [18]. In the future, this approach could be applied to other areas of HEAs research, providing theoretical support for experiments and further applications in structural applications, energy, therapeutics, and other fields [19–23].

2. Methodology

2.1. Analysis flow

Figure 1 shows the analysis flow for the present work. The data were generated using MD simulation. First, 1000 sets of variables are randomly generated, including different C and T. MD is used to obtain 1000 different models of HEAs samples. Each model undergoes tensile testing at various T to acquire stress–strain curves, which are used to calculate E and uT. Mechanical and phase characteristics of each sample group are computed based on elemental concentrations. Subsequently, a feature pool comprising 1000 rows and 62 columns, each containing 62 features, is constructed. Next, feature selection is performed using both Pearson Correlation Coefficient (PCC) and Principal Component Analysis (PCA). Separate ML models are constructed using the selected features for final training. The dataset was partitioned such that 80% was allocated for training, while the remaining 20% was equally divided, with 10% designated for testing and 10% reserved for validation. ML models such as XGBoost, AdaBoost, RF, LGBoost and DT are employed for prediction. Model optimization utilizes 10-fold cross-validation and grid search methods, with evaluation metrics including root mean squard error ($RMSE$), coefficient of determination (R^2), mean absolute error (MAE) and relative absolute error (RAE). Finally, the performance of selected features from both methods is compared during model training. The top 3 performing models and their corresponding parameters are identified. Shapley Additive Explanations (SHAP) technique is utilized to interpret the constructed models from both single-variable and multi-variable perspectives.

2.2. MD simulations and analysis

All simulations were performed using the open-source large-scale atomic/molecular massively parallel simulator (LAMMPS) [24]. The structural analysis and visualization were performed using OVITO [25]. The

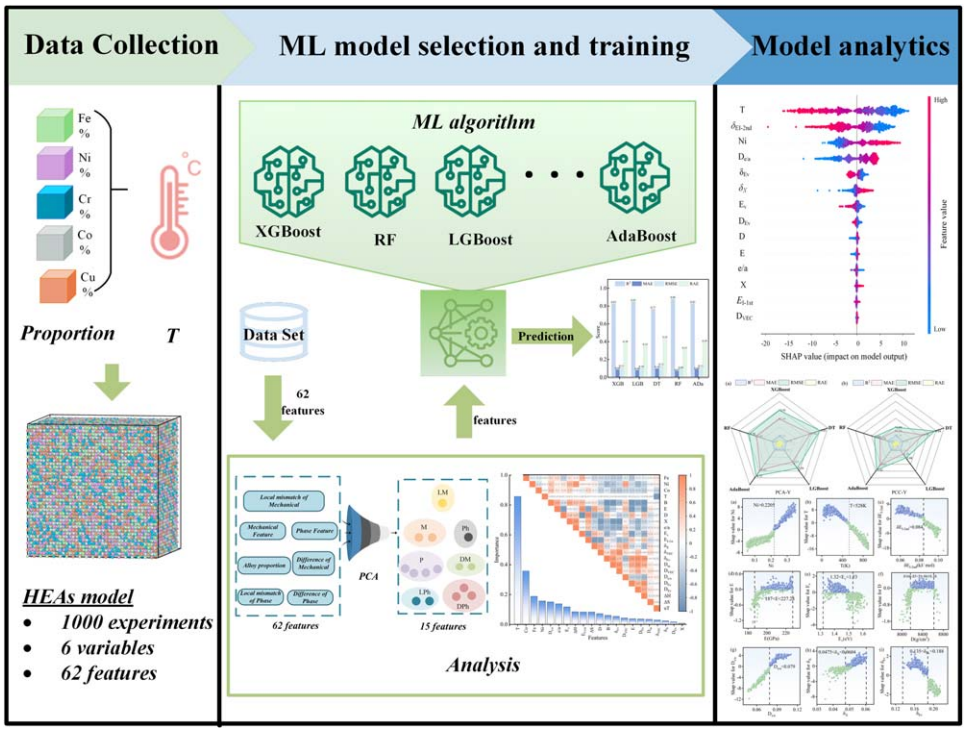

Figure 1. Schematic diagram showing the analysis flow in the present work.

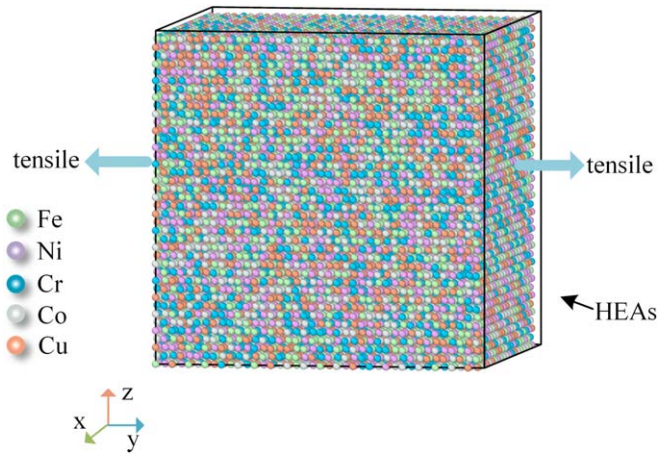

Figure 2. Simulation configurations.

HEAs models we used consist of single-phase face-centered cubic (FCC) FeNiCrCoCu HEAs, as shown in figure 2. The atomic configuration of HEAs starts with constructing a single-crystal Cu model, with an initial FCC lattice spacing of 3.62 Å [26], having dimensions of 73 Å × 73 Å × 73 Å. Subsequently, Cu atoms are replaced by other four types of atoms based on randomly generated atomic compositions, each ranging between 5 at% and 35 at%. Researchers have studied the deformation behavior of various types of HEAs across a wide T range from low to high T from −196 to 1000 °C, suggesting excellent mechanical properties over a broad T spectrum, thereby demonstrating significant potential for applications under harsh conditions [27–31]. Therefore, we set the T range from 25 K to 1000 K. The interactions between FeNiCrCoCu HEAs atoms are described using the embedded atom method (EAM) potential, which has been validated in previous studies on HEAs systems [6, 32–34]. The specific simulation process is outlined as follows:

In this study, the isothermal–isobaric (NPT) ensemble was employed throughout the simulation, and the conjugate gradient (CG) method was utilised to minimise the energy of the HEAs ensemble. The EAM potential function for the Fe-Ni-Cr-Co-Cu HEA, as provided by Diana Farkas *et al* [35], was adopted. In the simulation,

periodic boundary conditions are applied in all three directions, and the time step is set to 1 fs. The model is then relaxed for 100 ps to reach thermodynamic equilibrium, with the pressure in the x and z directions maintained at 0 bar. Tensile deformation is applied along the y-axis at different working temperatures, with a strain rate of $1 \times 10^9\ s^{-1}$, as it is a commonly used value in simulations of FeNiCrCoCu systems and other HEAs.

The MD's super-fast strain rate hinders the application of MD deformation simulations. Typical tensile strain rates of all-atomistic MD simulations are in the range of $10^8 \sim 10^9\ s^{-1}$, which is multiple orders of magnitude higher than tensile experiments [36]. Due to limitations in computational resources, simulations cannot achieve the same strain rates as in experiments, and a strain rate of $10^9\ s^{-1}$ is commonly used in simulations [37, 38]. As a result, the experimental results obtained from simulations may show some discrepancy from the real values, and the predictions of the ML model, which are based on MD simulations, tend to align more with theoretical values. Currently, some researchers have proposed solutions to address the discrepancy between MD and experimental strain rates, with the general approach being to establish a connection between strain rate and mechanical properties [39, 40].

The computational resources employed in this study are based on a high-density simulation server comprising 80 nodes interconnected via a 200 G optical fibre, totalling 4480 CPU cores, with a theoretical computational peak performance of 89 trillion operations per second. Additionally, a high-density computing server with 1152 cores, also based on optical interconnection, serves as a backup resource. Consequently, the existing experimental equipment and conditions are fully capable of meeting the requirements of this research.

2.3. Calculation and dimensionality reduction of characteristic parameters

The values of E and uT are determined by the stress–strain curve. The value of E is obtained using the least squares method, with the following fitting equation:

$$\hat{b} = \frac{\sum_{i=1}^{n} x_i y_i - n\overline{x}\overline{y}}{\sum_{i=1}^{n} x_i^2 - n\overline{x}^2} \tag{1}$$

where x_i represents the strain value for the i_{th} data point, y_i represents the stress value for the i_{th} data point, and $\hat{b}$ represents the E.

The parameters used in this study include intrinsic atomic properties and certain empirical parameters established in previous research [41]. We employed various computational formulas and existing parameters to derive a total of 62 features. The elemental basic parameters necessary for these features are derived from established research outcomes [42, 43]. The feature pool comprises 30 descriptors, with detailed information presented in table 1.

In table 1, 56 features are computed, where n represents the number of elements in the alloy, which is 5 in the current study. f_i denotes the basic parameters of the i_{th} element, c_i represents the concentration of the element, and the Packing misfit parameter formula includes R, which denotes the Boltzmann constant, and $\Delta H_{ij}^{\mathrm{mix}}$, representing the enthalpy correlation between two different elements. Detailed descriptions of these parameters are obtained from peer-reviewed papers [41, 44]. Additionally, concentrations of the five components in HEAs and T are included in the feature pool, comprising a total of 62 feature descriptors in the feature pool.

After constructing the feature pool, both PCC and PCA methods were utilized to select important features. PCA is employed to analyze and simplify data [45], widely applied across various research fields [46–48]. Its fundamental principle involves identifying a small number of principal component variables from extensive data, reducing dimensionality while retaining as much original information of the dataset as possible [49]. On the other hand, PCC is used to calculate the correlation between any two feature descriptors, with its computation results ranging between −1 and 1. In this study, for the PCA method, we utilized it for dimensionality reduction, a process implemented using Scikit-Learn. The minimum proportional threshold that represents the variance sum of the principal components is 0.95, i.e. the value of 'n_components' is 0.95. Based on the characteristics of the dataset, we divided the data into seven categories: alloy ratio, phase characteristics, mechanical characteristics, the difference between mechanical and phase characteristics, and local mismatch values, and performed dimensionality reduction seven times. For the PCC method, we first calculated the correlation between any two features, then filtered out all features where the absolute value of the correlation between a certain feature and other features was greater than 0.95, retaining features with higher correlation to the results.

2.4. Model training and evaluation methods

In the context of predicting the mechanical properties of HEAs, five commonly used ML algorithms — XGBoost, AdaBoost, LGBoost, DT, and RF — are considered. In order to optimise the model, the grid search and k-fold cross-validation methods are used, with k discounted by 10 percent. This is done in order to address the issue of minimizing overfitting [50]. In order to evaluate the model, four metrics have been selected: *RMSE*,

Table 1. Detailed description and formula used to calculate each feature.

Description	Abbreviation	Formula
G	Shear modulus	$F = \sum_{i=1}^{n} C_i f_i$
B	Bulk modulus	
E	Young's modulus	
R	Radius	
v	Poisson's ratio	
D	Density	
TB	Boiling point	
m	Relative Atomic Mass	
X	Pauling electronegativity	
T_m	Melting point	
VEC	Valence electron concentration	
e/a	Free electron concentration	
E_c	Cohesive energy	
E_v	Vacancy formation energy	
$E_{\text{I-1st}}$	First ionization energy	
$E_{\text{I-2nd}}$	Second ionization energy	
δ_F	Differences in weighting criteria for basic parameters	$\delta_F = \sqrt{\sum_{i=1}^{n} C_i \left(1 - \frac{f_i}{F}\right)^2}$
D_F	Local mismatch value of basic parameters	$D_F = \sum_{i=1}^{n} \sum_{j=1, i \neq j}^{n} C_i C_j \lvert f_i - f_j \rvert$
ΔS	Configurational entropy	$\Delta S = -R \sum_{i=1}^{n} C_i \ln (C_i)$
ΔH	Mixing enthalpy	$\Delta H = \sum_{i=1, j>i}^{n} 4 C_i C_j \Delta H_{ij}^{\text{mix}}$
Ω	Phase stability descriptor	$\Omega = T_m * \frac{\Delta S}{\lvert \Delta H \rvert}$
Γ	Packing misfit parameter	$\Gamma = \frac{1 - \sqrt{\frac{(R + R_{\min})^2 - R^2}{(R + R_{\min})^2}}}{1 - \sqrt{\frac{(R + R_{\max})^2 - R^2}{(R + R_{\max})^2}}}$
Λ	Λ parameter	$\Lambda = \Delta S_{\text{mix}} / \delta R^2$

R^2, MAE and RAE. The following definitions are provided for each of the aforementioned terms:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{2}$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \tag{3}$$

$$MAE = \frac{1}{n} \sum_{i=1}^{n} \lvert y_i - \hat{y}_i \rvert \tag{4}$$

$$RAE = \frac{\sum_{i=1}^{n} \lvert \hat{y}_i - y_i \rvert}{\sum_{i=1}^{n} \lvert \bar{y} - y_i \rvert} \tag{5}$$

where n represents the number of observations, y_i represents the true value, $\hat{y}_i$ represents the predicted value of the model, and $\bar{y}$ represents the average value.

3. Results and discussion

3.1. Dimensionality reduction using PCA

In this study, PCA was utilized to perform dimensionality reduction on the features. As depicted in figure 3, the left side illustrates the classification of features in the feature pool before reduction, categorized into seven types based on different calculations of mechanical and phase characteristics, totaling 53 features. The numbers in parentheses within each category indicate the current count of features. The right side shows the remaining number of features after reduction, resulting in a total of 15 features across the seven categories. This selection is based on the variance explained ratio derived from PCA. The cumulative variance proportion threshold was set at 0.95 in order to ensure that 95% of the original data's information content is retained. The PCA algorithm automatically identifies the minimum number of features required to meet this criterion, ultimately determining that 15 features should be retained. Features such as T, ΔS and ΔH that are not suitable for reduction were excluded from this process.

Post dimensionality reduction, a new dataset was reconstructed containing 22 features, detailed in table 2. Features labeled (16)–(22) denote those that were not reduced, with the subscript numbers indicating their relative importance within their respective attributes, where lower numbers signify higher importance.

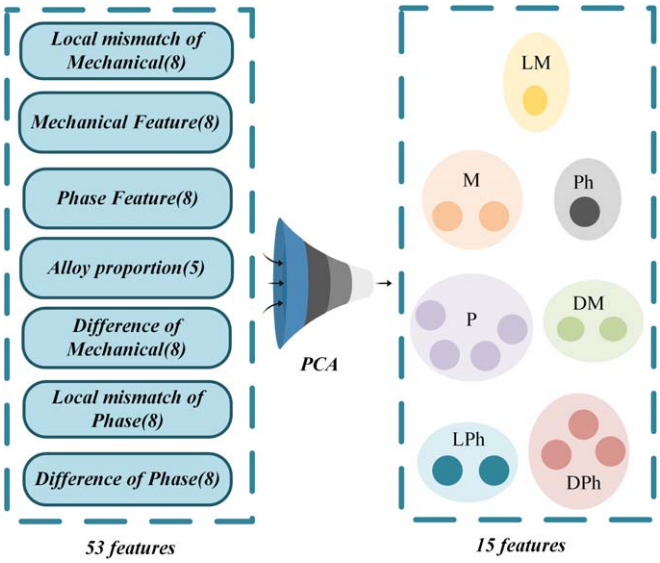

Figure 3. Results of dimensionality reduction of seven types of features by PCA.

Table 2. Elemental base parameters.

Serial number	Abridge	Full name
(1)	P_0	Principal components of alloy proportion
(2)	P_1	
(3)	P_2	
(4)	P_3	
(5)	M_0	Principal components of mechanical features
(6)	M_1	
(7)	Ph_0	Principal components of phase features
(8)	DM_0	Difference of principal components of mechanical features
(9)	DM_1	
(10)	DPh_0	Difference of principal components of phase features
(11)	DPh_1	
(12)	DPh_2	
(13)	LM_0	Local mismatch of principal components of mechanical features
(14)	LPh_0	Local mismatch of principal components of phase features
(15)	LPh_1	
(16)	T	Simulated temperature
(17)	ΔH	Mixing enthalpy
(18)	ΔS	Configurational entropy
(19)	$\delta_{\Delta H}$	Difference in mixing enthalpy
(20)	Ω	Phase stability descriptor
(21)	Γ	Packing misfit parameter
(22)	Λ	Λ parameter

Specifically, among the alloy proportion features, 4 out of 5 were retained; for mechanical performance features, 2 out of 8 were retained; phase characteristics were reduced to 1 out of 8; local mismatch features of mechanical and phase characteristics were reduced to 1 and 2 out of 8, respectively; and differences in mechanical and phase characteristics were reduced to 2 and 3 out of 8, respectively.

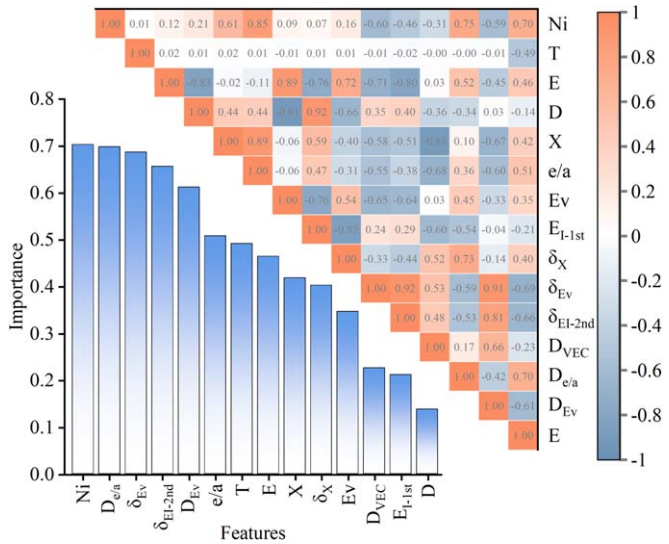

Figure 4. In order to predict *E*, the heat map of the remaining 14 features is calculated using PCC, following the process of feature screening. The degree of correlation is indicated by the colour intensity. The illustration in the lower left-hand corner depicts the relative importance of the 14 features.

3.2. Feature filtering using PCC

In ML, PCC is a widely used method for feature selection, extensively employed by many researchers [51, 52]. Figure 4 displays a heatmap and feature importance ranking after undergoing PCC filtering, used to predict the *E* of HEAs. In the top right corner of the figure, 14 features remain after PCC filtering from the initial 62 features in the feature pool, showing their correlations with each other. Darker shades of orange indicate positive correlations, darker shades of blue indicate negative correlations, and white denotes no correlation. The inset in the bottom left corner illustrates the importance ranking of each feature. Notably, parameters such as *Ni* (representing the proportion of *Ni* in the HEAs), $D_{e/a}$, δ_{Ev}, $\delta_{EI\text{-}2nd}$, and D_{Ev} are highly correlated features, with absolute correlations to *E* exceeding 0.6. All selected features have absolute correlations with *E* greater than 0.1.

In the evaluation of predicting *E* using five common ML models, XGBoost emerged as the most effective model. Figure 5 presents the analysis of feature importance in the XGBoost model's prediction of *E*. Figure 5(a) shows the importance ranking of the 14 features, where *T* is identified by the XGBoost model as the most crucial feature, with *T* and *Ni* significantly more important than other features. Figures 5(b) and (c) illustrate the impacts of *Ni* and *E* (calculated according to table 1) on *E*. In these figures, orange indicates *T* above 300 K, while green indicates *T* below 300 K. At *T* below 300 K, HEAs may exhibit higher *E* values. These insights from the XGBoost model highlight the influential role of *T* and *Ni* content in determining *E* in HEAs, suggesting *T* as a significant factor particularly affecting the material's mechanical properties at lower *T*.

Figure 6 shows the heatmap and feature importance ranking after PCC filtering, used to predict the *uT* of HEAs. In the top right corner, 21 features remain after PCC filtering from the initial 62 features, displaying their correlations. Darker colors indicate higher correlations, while lighter colors indicate lower correlations. The feature *T* has the highest importance. Elements *Co*, *Fe*, and *Ni*, representing their proportions in the alloy, indicate that the alloy composition is crucial for *uT*. Among the 21 selected features, 19 features have correlations lower than 0.2, emphasizing the significant impact of *T* and *C* on predicting the *uT* of HEAs.

3.3. Optimal model and results

The parameter tuning process for XGBoost involved optimizing the 'max_depth', 'learning_rate' and 'n_estimators' parameters. Based on experience, the tuning range for 'max_depth' is [3, 25]. The tuning range for 'n_estimators' is [5,200] with a step size of 5. The tuning range for 'learning_rate' is [0.1,1.1] with a step size of 0.2. The parameter tuning process for LGBoost involved optimizing the 'max_depth' and 'n_estimators' parameters. The tuning range for 'max_depth' is [1, 31] with a step size of 5. The tuning range for 'n_estimators' is [10,101] with a step size of 10. The parameter tuning process for AdaBoost involved optimizing the 'learning_rate' and 'n_estimators' parameters. The tuning range for 'learning_rate' is [0.1,1.1]. The tuning range for 'n_estimators' is [5,200]. The parameter tuning process for DT involved optimizing the ' min_samples_split' and 'min_samples_leaf' parameters. The tuning range for 'min_samples_split' is [2, 10] with a step size of 1. The tuning range for 'min_samples_leaf' is [2, 5] with a step size of 1. The parameter tuning process for RF involved

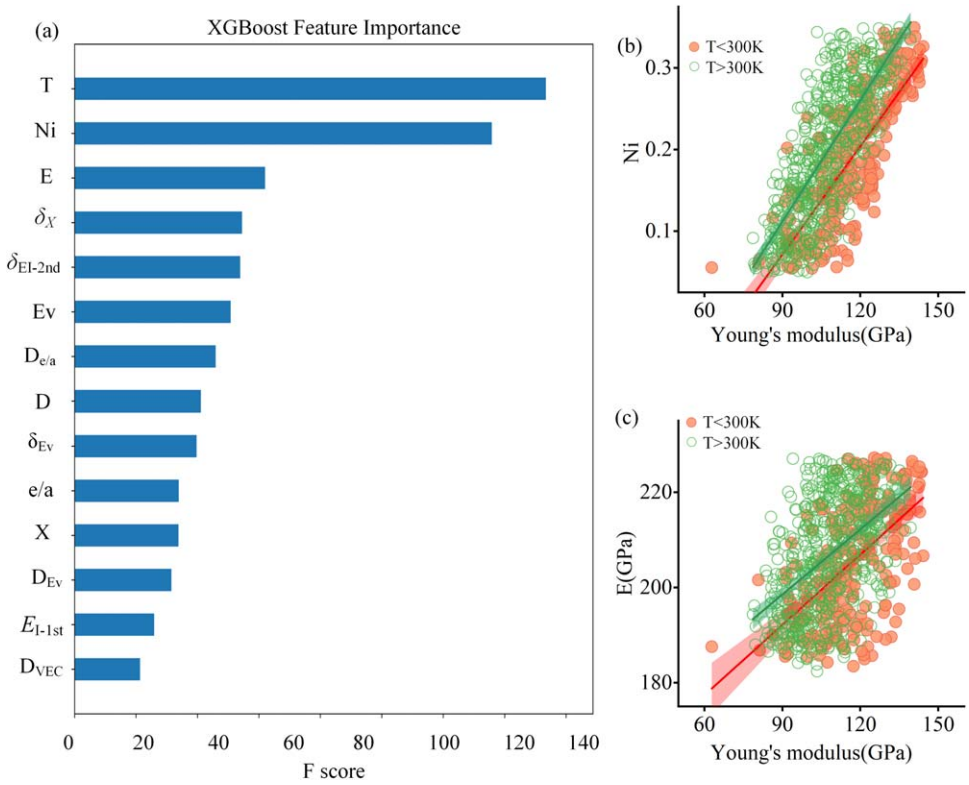

Figure 5. Feature importance analysis for predicting *E* by XGBoost algorithm. (a) Feature importance ranking when using XGBoost model (b) Influence of *Ni* and *T* on *E* (c) Influence of *E* (calculated according to table 1) and *T* on *E* (calculated according to Stress–strain curve). Orange indicates T > 300 K, and green indicates T < 300 K.

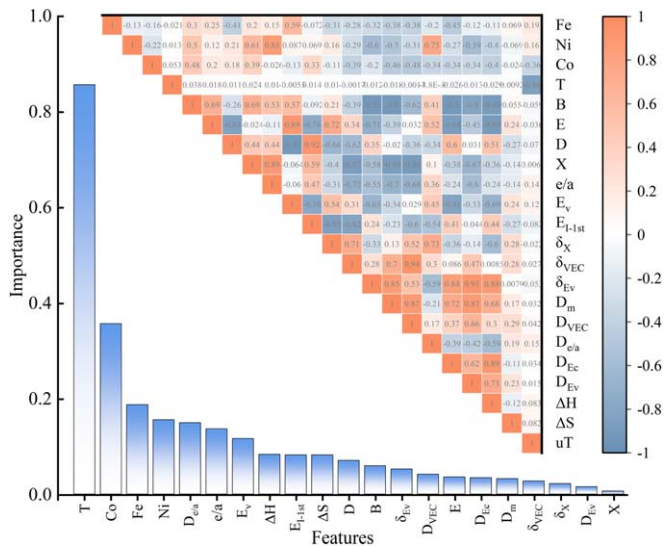

Figure 6. The heat map after feature screening is used by PCC When predicting *uT*. The darker the color, the higher the correlation, and the lighter the color, the lower the correlation.

optimizing the 'max_depth' and 'n_estimators' parameters. The tuning range for 'max_depth' is [1, 51] with a step size of 5. The tuning range for 'n_estimators' is [10,101] with a step size of 10.

Whether features were selected using PCA or PCC, the performance of the five ML models on the training set was generally satisfactory, with R^2 values exceeding 0.8. The predictive performance of these models on the test set is crucial [41]. Figure 7 illustrates the performance of the five ML models using features selected by PCC and PCA methods. Figures 7(a) and (b) are radar charts comparing the prediction performance for *E* of HEAs using the two feature selection methods. Four evaluation metrics are used to assess the ML models' performance, each

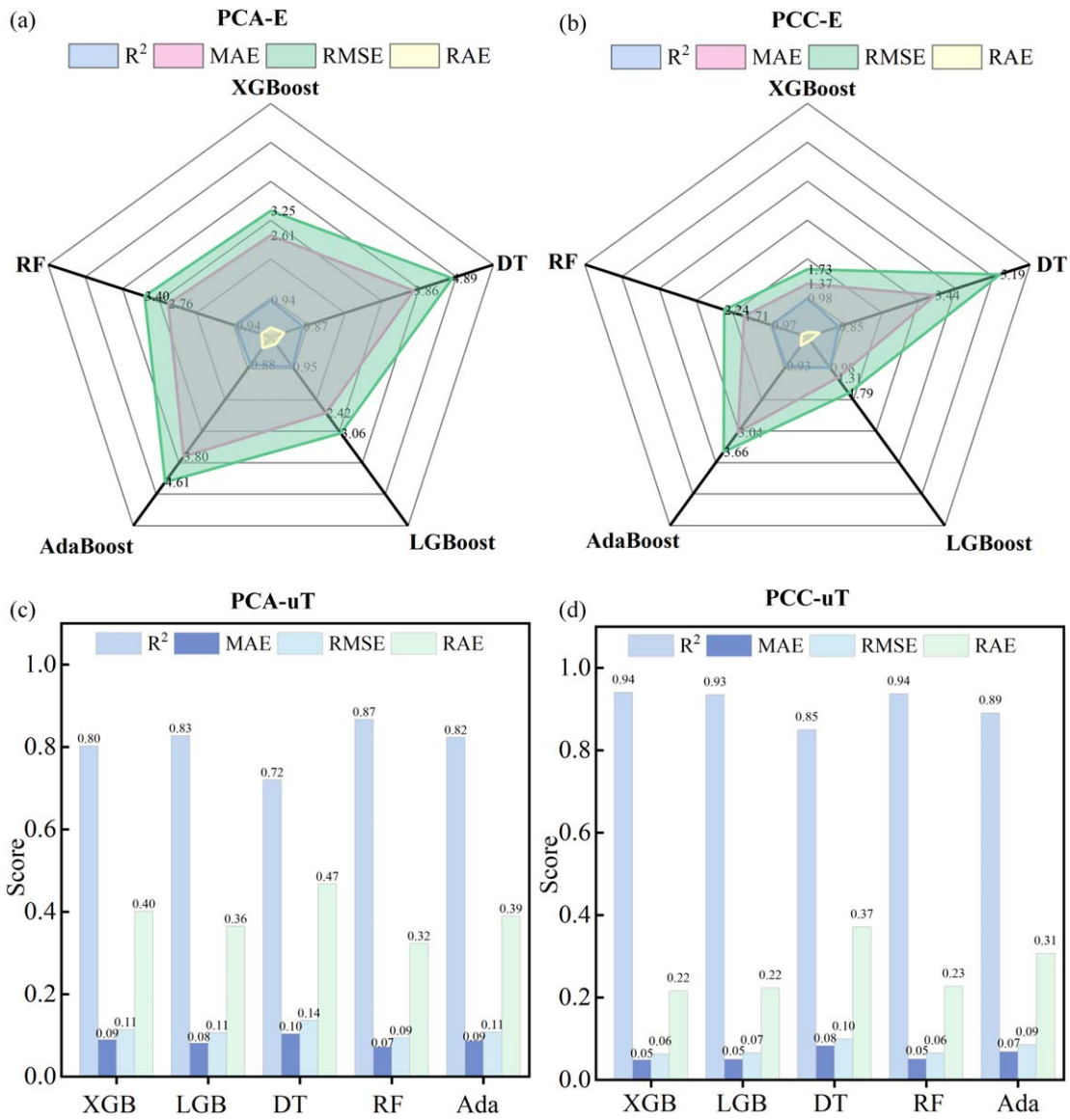

Figure 7. A comparative analysis of the performance of five ML models under different feature screening methods. (a) The radar map compares the prediction accuracy and error of five ML models on the *E* test set after using PCA to filter features. (b) The radar map compares the prediction accuracy and error of five ML models on the *E* test set after the use of the PCC method to filter features. (c) Prediction of *uT* by five ML models following the application of PCA method to identify pertinent features. (d) The prediction of *uT* by five ML models after screening features using PCC method.

represented by a different color. Generally, for R^2, values closer to 1 indicate better model fit. Conversely, for *MAE*, *RMSE*, and *RAE*, values closer to 0 indicate better model fit. Comparing the two radar charts, figure 7(a) shows larger areas for *MAE*, *RMSE*, and *RAE*, and a smaller area for R^2 compared to figure 7(b). This indicates that features selected by the PCC method performed better in the training set for predicting *E*. Although the PCA method did not predict *E* as well as PCC, XGBoost, RF, and LGBoost achieved R^2 values greater than 0.9, and all five models had R^2 values greater than 0.85. This demonstrates that PCA is a viable method, particularly when dealing with a large number of features, as its selection process is simpler compared to PCC.

Figures 7(a) and (b) feature bar charts comparing the prediction performance for HEAs *uT* using the two feature selection methods. The same four evaluation metrics are used to assess the ML models' performance, each represented by a different color. The highest R^2 values were achieved by XGBoost ($R^2 = 0.94$) and RF ($R^2 = 0.94$) models using PCC for feature selection. The lowest R^2 was observed with the DT model using PCA ($R^2 = 0.72$), indicating that XGBoost and RF with PCC performed best on the training set. The highest *MAE* was found in the DT model with PCA ($MAE = 0.10$), while the lowest *MAE* was observed in the XGBoost, LGBoost, and RF models using PCC ($MAE = 0.05$). Similarly, the highest *RMSE* was in the DT model with PCA ($RMSE = 0.14$), and the lowest *RMSE* was found in XGBoost and RF models using PCC ($RMSE = 0.06$). The highest *RAE* occurred in the DT model with PCA ($RAE = 0.47$), while the lowest *RAE* was in the XGBoost ($RAE = 0.22$) and LGBoost ($RAE = 0.22$) models using PCC. Overall, features selected by the PCC method

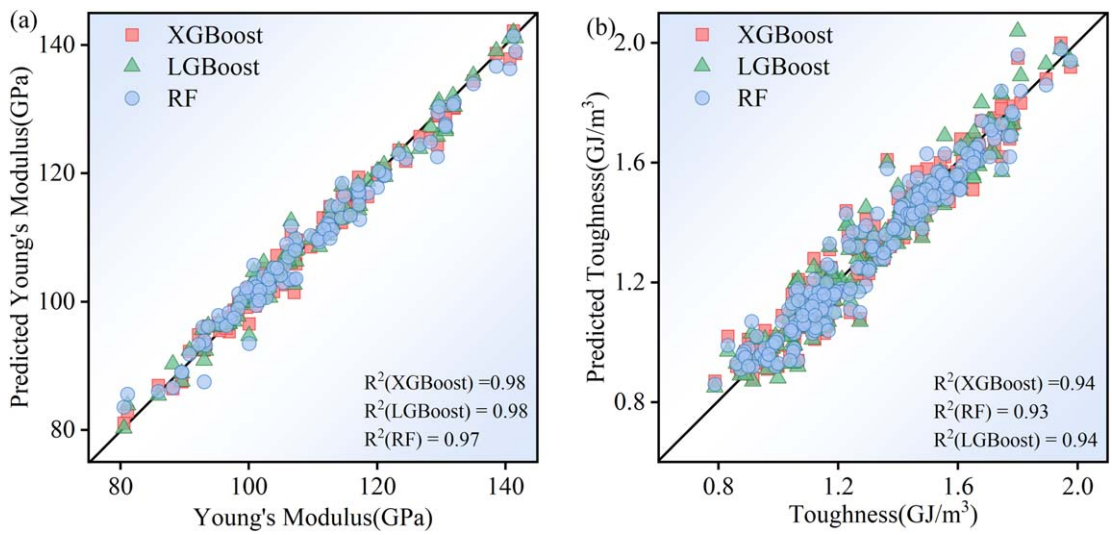

Figure 8. A comparative analysis of the results obtained from the prediction of mechanical properties utilising the PCC feature screening method. (a) The XGBoost, LGBoost and RF models were used to predict *E*. (b) Using XGBoost, LGBoost, and RF models to predict *uT*. The predictions of all models are represented by different colours.

demonstrated better performance on the test set, with four models achieving R^2 values greater than 0.9. Among the considered ML models, the XGBoost model achieved the highest regression prediction accuracy, both for predicting *E* and *uT*. This indicates that the XGBoost model shows superior performance in regression predictions of *E* and *uT* and has stronger generalization capability compared to other ML models.

Based on the analysis of figure 7, XGBoost, LGBoost, and RF were identified as the top-performing models. Figure 8 illustrates the different performances of these three models on the test set using the PCC feature selection method. In this figure, the same color represents the predicted values of different test points within the same model, while different colors correspond to different ML models. Data points closer to the diagonal line indicate better model predictions. From figure 8(a), it is evident that all three models have R^2 values greater than 0.9, with all test data predictions concentrated around the diagonal line. This indicates that the models perform well in predicting *E*. Figure 8(b) shows the prediction performance for *uT* by the three models. Although the performance is not as strong as in figure 8(a), the R^2 values for all models are still above 0.8. This demonstrates that while the prediction accuracy for *uT* is slightly lower than for *E*, the models still achieve a good level of accuracy, making XGBoost, LGBoost, and RF effective for both predictions.

As illustrated in figure 9, the learning curves of the XGBoost model trained using features selected by the PCC method are presented. The horizontal axis of the figure represents the number of iterations, while the vertical axis indicates the variation in loss values. As depicted in the figure, both the training loss and validation loss exhibit a declining trend and eventually stabilise as the training progresses. This demonstrates favourable convergence behaviour, with no significant signs of overfitting or underfitting.

Figure 10 presents the performance of the three models (XGBoost, LGBoost, and RF) on the test set using the PCA feature selection method. Data points closer to the diagonal line indicate better model predictions. From figure 10(a), which shows the performance of the three models in predicting *E*, it is clear that all three models have R^2 values greater than 0.9. The predicted values for all test data are concentrated around the diagonal line, indicating that the models perform well in predicting *E*. Figure 10(b) illustrates the prediction performance for *uT* by the three models. Although the performance is not as strong as in figure 8(a), the R^2 values for all models are still above 0.8. This shows that while the prediction accuracy for *uT* is slightly lower than for *E*, the models still achieve a good level of accuracy. Therefore, XGBoost, LGBoost, and RF remain effective for predicting both *E* and *uT*, even when using the PCA feature selection method.

Figure 11 illustrates the learning curves of the LGBoost and RF models trained using features selected by the PCA method. The horizontal axis represents the number of iterations, while the vertical axis indicates the variation in loss values. As shown in the figure, both the training loss and validation loss exhibit a declining trend and eventually stabilize as the training progresses. However, compared to the learning curves of the PCC method, the performance of the PCA method is inferior, particularly on the validation set.

Overall, the prediction performance for *E* on the test set is better than for *uT*, with R^2 values greater than 0.9 for all three models, regardless of whether PCC or PCA methods were used. For *uT* predictions, the R^2 values for all three models are above 0.8. The features selected using the PCC method performed better on the test set compared to those selected using the PCA method. In terms of the convergence of learning curves on both the

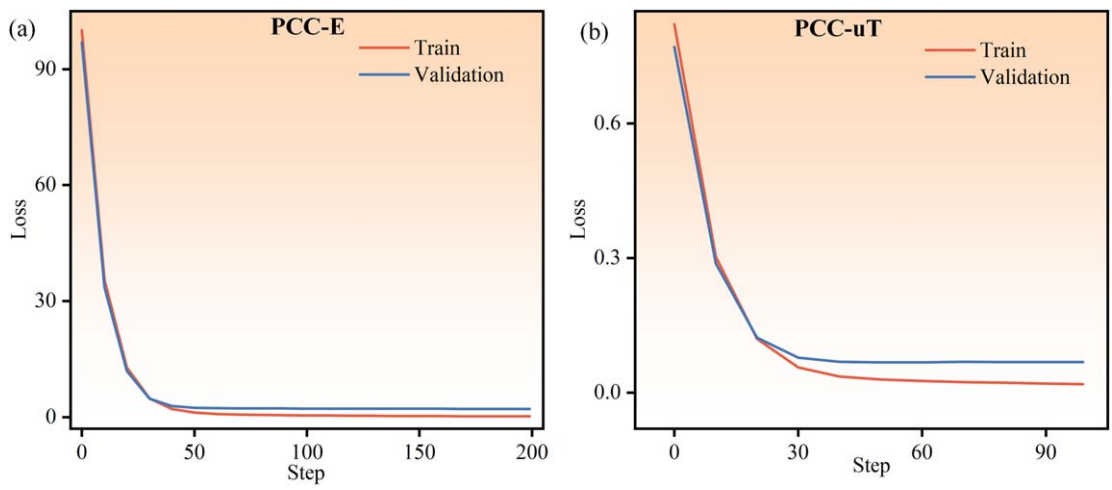

Figure 9. Loss variation during the training of the ML model after feature selection using PCC. (a) The loss variation with increasing training iterations for the XGBoost model predicting *E*. (b) The loss variation with increasing training iterations for the XGBoost model predicting *uT*.

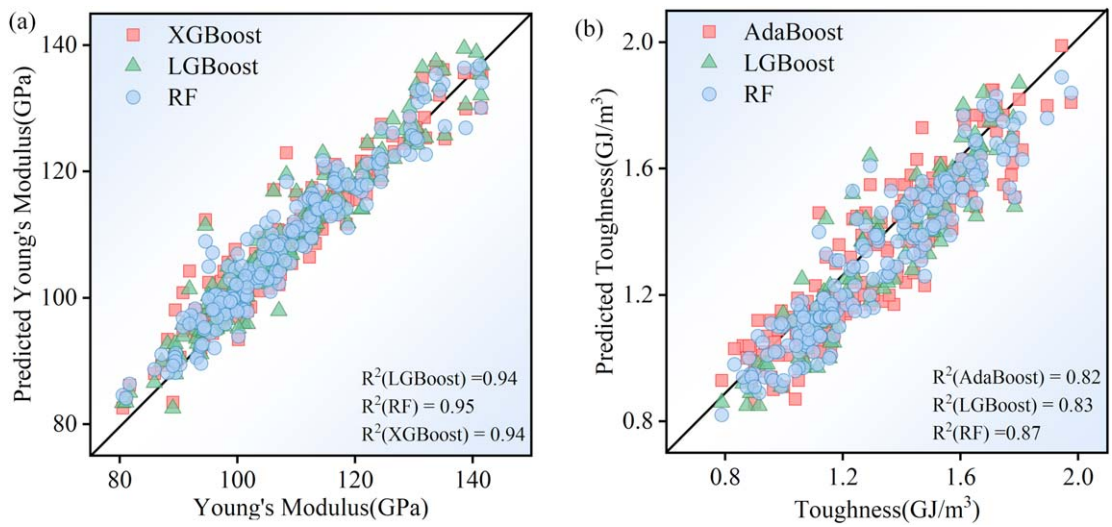

Figure 10. A comparison of the results of predicting mechanical properties using the PCA feature screening method. (a) The XGBoost, LGBoost and RF models were used to predict *E*. (b) The XGBoost, LGBoost, and RF models were employed to predict *uT*. The predictions of all models are represented by different colours.

training and validation sets, the PCC method demonstrates more desirable performance. XGBoost and RF models generally outperformed the other models. Therefore, subsequent analyses will focus primarily on using the PCC method for feature selection and choosing XGBoost as the research model.

3.4. Model explanation

The interpretation of ML models is crucial, and SHAP provides opportunities for flexible modeling, explanation, and visualization of complex phenomena and processes [53]. In this study, the SHAP method was used to interpret the XGBoost model. Figure 12 presents a global SHAP feature importance plot constructed during the prediction of *E* using the XGBoost model. It describes the impact of each feature on the model output and ranks the features based on their SHAP values. Red indicates high feature values, blue indicates low feature values, and the horizontal axis represents SHAP values, where positive values indicate a positive impact on the prediction and negative values indicate a negative impact. From figure 12, it is evident that *T* is the most important feature in the XGBoost model's predictions. As *T* increases, the *E* of HEAs tends to decrease, showing a negative correlation. This is because lower *T* reduce atomic or molecular vibrations, increasing the material's stiffness. Conversely, at higher *T*, the *E* typically decreases due to increased thermal vibrations of atoms or molecules, making the material more prone to deformation. *Ni*, $D_{e/a}$ and δ_X are positively correlated with the *E*

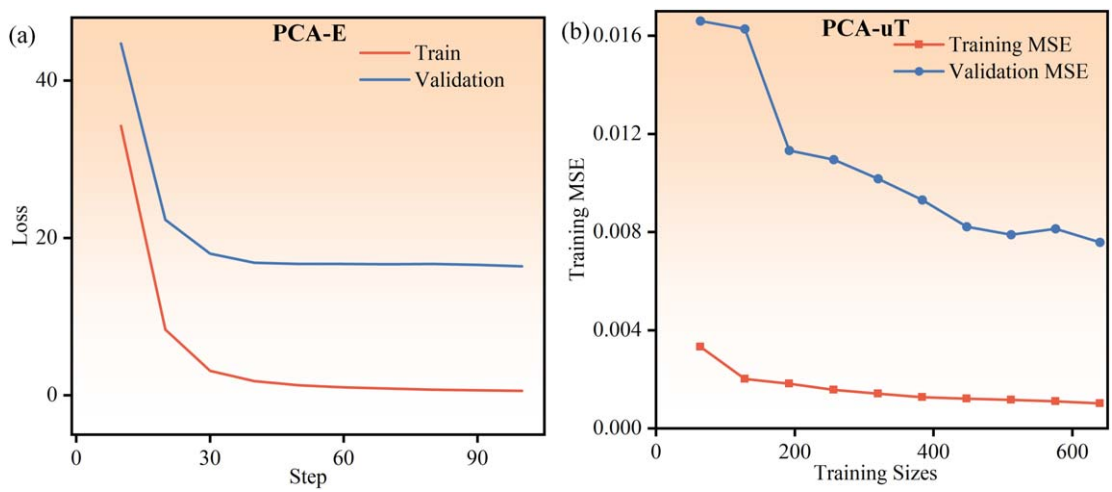

Figure 11. Loss variation during the training of the ML model after feature selection using PCA. (a) The loss variation with increasing training iterations for the LGBoost model predicting *E*. (b) The loss variation with increasing training iterations for the RF model predicting *uT*.

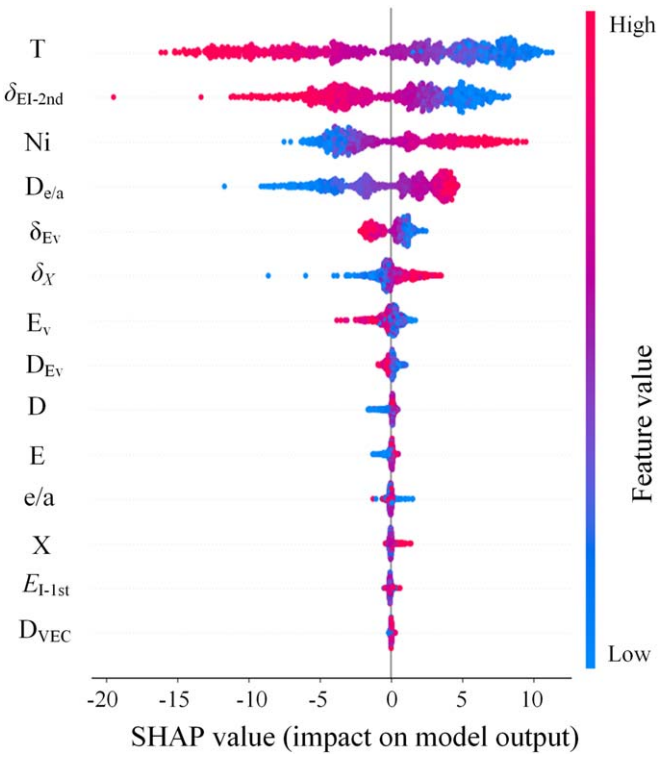

Figure 12. Feature importance analysis of XGBoost algorithm.

of HEAs, meaning that as their values increase, the *E* also increases. This indicates that these features contribute positively to the stiffness and strength of the material.

As shown in figure 13, the analysis of individual features and model output is depicted, where the horizontal axis represents the feature values and the vertical axis represents the SHAP values for each feature. Based on the feature importance ranking from figure 13, nine features were selected for study: T, $\delta_{\text{EI-2nd}}$, Ni, $D_{e/a}$, δ_{Ev}, δ_X, E_v, D_{Ev} and D. Each feature's critical value approximately divides the SHAP values into positive and negative regions. A SHAP value greater than 0 indicates that increasing the feature will increase the *E*. Therefore, to enhance the *E*, we suggest the following critical conditions for each feature: $T<528$ K, $\delta_{\text{EI-2nd}} > 0.084$(kJ mol^{-1}), $Ni > 0.2205$, $D_{e/a} > 0.079$, $0.135 < \delta_{Ev} < 0.188$, $\delta_X > 0.0475$, $1.32(\text{eV}) < E_v < 1.53(\text{eV})$, $D_{Ev} < 0.294(\text{eV})$ and $8166.85(\text{g cm}^{-3}) < D < 8613.38(\text{g/cm}^3)$.

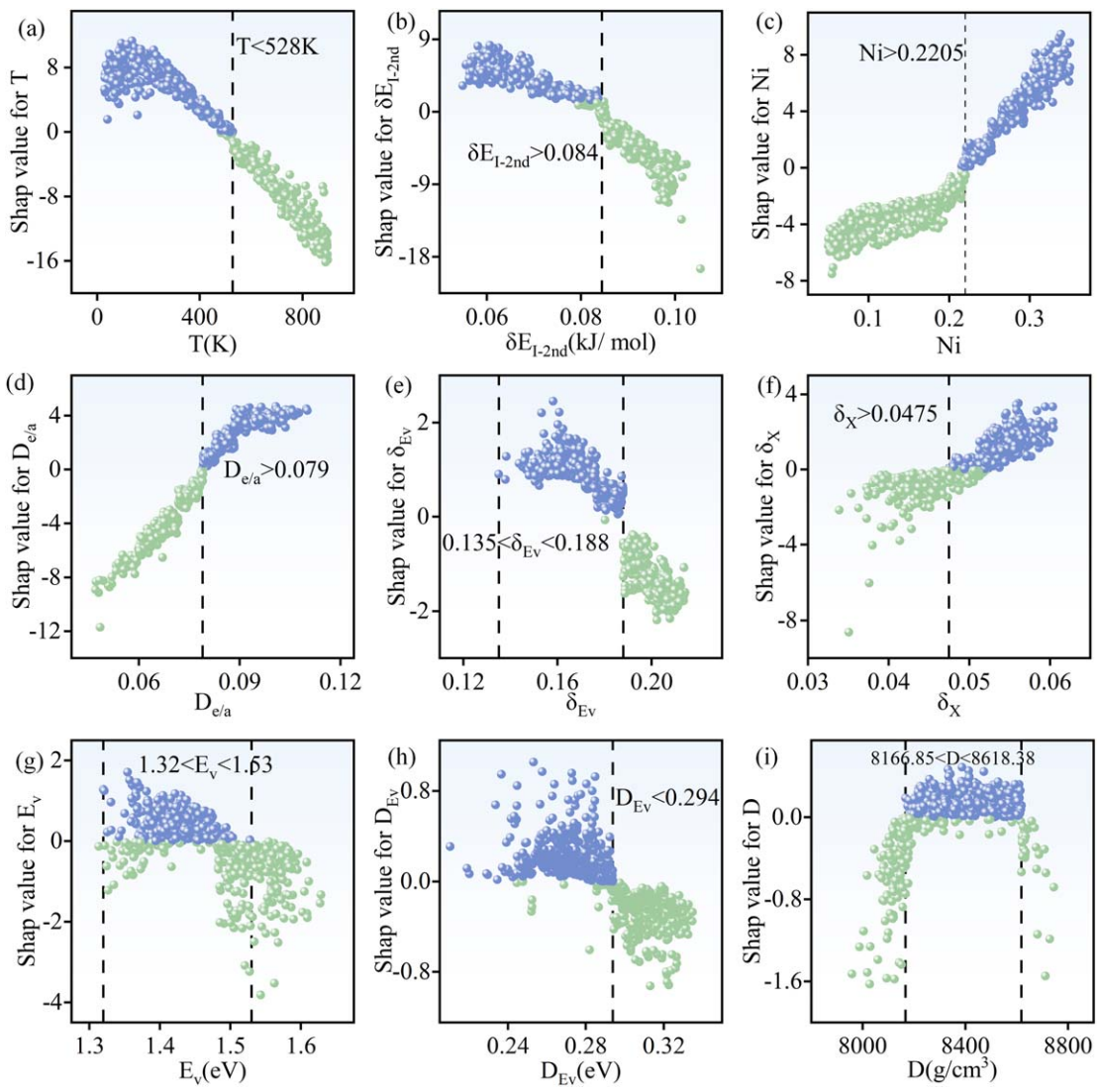

Figure 13. The local dependence of each feature in the ML model. (a) T (b) $\delta_{\mathrm{EI\text{-}2nd}}$, (c) Ni (d) $D_{e/a}$, (e) δ_{Ev}, (f) δ_X, (g) E_v, (h) D_{Ev} and (i)D.

The time and spatial scales of MD simulations are limited, typically only capable of simulating time frames on the nanosecond scale and spatial ranges on the nanometer scale. The strain rate used in this study is $1 \times 10^9\ \mathrm{s}^{-1}$, which is significantly higher than the strain rates under experimental conditions. This discrepancy may lead to differences between the simulation results and the actual experimental observations.

4. Conclusion

This study utilized C and T as initial variables, combining MD simulations with ML to predict the mechanical properties of FeNiCrCoCu HEAs, specifically E and uT. MD simulations provided stress–strain curves and mechanical properties for 1000 HEAs samples, and 55 derived features were calculated, forming an original dataset with 62 features. Two feature selection methods, PCA and PCC, were employed. PCA primarily performed feature dimensionality reduction, while PCC focused on selecting relevant features. Both methods aimed to identify useful features from the feature pool. Five ML models were used to predict the two mechanical properties of HEAs. Techniques such as 10-fold cross-validation, grid search, and others were employed during model training to determine the best models and parameters. The SHAP method was used to explore the impact of features on performance, explaining the importance of each feature to the mechanical properties of HEAs.

The results indicate that the PCC feature selection method is superior to PCA, whether for predicting E or uT of HEAs. E performed best on the test set, with four out of five models achieving an R^2 greater than 0.9. XGBoost outperformed other models, and SHAP was employed to explain the importance of each feature in the XGBoost model. Features T, $\delta_{\mathrm{EI\text{-}2nd}}$, Ni and $D_{e/a}$ significantly influence the mechanical performance of HEAs. T exhibits a negative correlation with E, indicating that lowering T enhances mechanical performance, while increasing T

has a suppressive effect on *E*. This is attributed to the increased activity of dislocation movement within the material at elevated temperatures, which may lead to a higher defect density and, consequently, a reduction in Young's modulus. This study may provide a new method for enhancing the mechanical properties of HEAs through the combination of ML and MD, thereby uncovering their potential value and guiding experimental work.

Acknowledgments

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Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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